

DEYUAN (MIKE) HE

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🏛️ EDUCATION

🏠 PRINCETON UNIVERSITY, Princeton, NJ 2022 – 2027 (expected)

Ph.D. in Computer Science · Advisor: Prof. Aarti Gupta

Research Interests: Formal Methods; Coding Agents; Distributed and Agentic Systems

🏠 University of Washington, Seattle, WA 2018 – 2022

B.S. in Computer Science

Advisors: Prof. Zachary Tatlock & Dr. Steven Lyubomirsky

📖 SELECTED PUBLICATIONS (full list on [Google Scholar](#))

- **SOTERIA: Formally Verified Planning with Runtime Enforcement for Safe LLM Agents**
Mike He, A. Desai, S. Malik, A. Gupta. *Under Review*, 2026.
- **SPECY: Learning Specifications for Distributed Systems from Event Traces**
Mike He, A. Desai, A. Jagarapu, D. Terry, S. Malik, A. Gupta. *OOPSLA*, 2026.
- **ALGOVERI: An Aligned Benchmark for Verified Code Generation on Classical Algorithms**
H. Zhao*, Z. Yang*, J. Li*, Mike He*, Z. Li*, C. Jin, V. Veeravalli, A. Gupta, S. Arora. *ICML*, 2026. (Spotlight)
- **Goedel-Code-Prover: Hierarchical Proof Search for Open State-of-the-Art Code Verification**
Z. Li*, Z. Yang*, Mike He, H. Zhao, A. Zhao, S. Tang, K. Yang, A. Gupta, Z. Su, C. Jin. *COLM*, 2026.
- **Theory-Level Autoformalization: From Isolated Statements to Unified Formal Knowledge Bases**
M. J. Min, Mike He, Z. Li, Z. Yi, S. Malik, A. Gupta, X. Si, O. Bastani. *ICML*, 2026. (Spotlight)
- **Dynamic Tensor Rematerialization**
M. Kirisame*, S. Lyubomirsky*, A. Haan*, J. Brennan, Mike He, et al.. *ICLR*, 2021. (Spotlight)
- **Ranking Formal Specifications using LLMs**
Mike He, Z. Ang, A. Desai, A. Gupta. *LMPL Workshop*, 2025.
- **A Benchmark of Expert-Level Academic Questions to Assess AI Capabilities**
Center for AI Safety, Scale AI, and HLE Consortium. *Nature*, 2026. (Top-10 late contributor, CS/AI)

*: equal contribution.

🔍 RESEARCH EXPERIENCE

Princeton University, Princeton, NJ 2022 – Present

Graduate Research Assistant

- **SOTERIA** — verified planning and runtime enforcement for LLM agents (*Python*; 2026)
Built: an agentic framework for structured, hierarchical planning — a Python-like DSL expressing composable sub-plans, a verification-condition generator checking temporal and relational data-flow properties, and a runtime enforcement layer binding execution to the plan.
Result: **zero policy violations** over **207 tasks** (τ^2 -bench, Toolathon) across 4 LLM-backed agents.
- **ALGOVERI** — vericoding benchmark for classical algorithms (*Dafny*, *Verus*, *Lean 4*; 2026)
Built: curated aligned formal specifications for **77 classical algorithms** across all three languages, yielding a cross-language benchmark for verified code generation.
Result: best-model success collapses across languages — **40.3%** (*Dafny*), **24.7%** (*Verus*), **7.8%** (*Lean 4*) — isolating verification-language design as the dominant factor.
- **Goedel-Code-Prover** — LLM-guided proof search for code verification (*Lean 4*; 2026)
Built: curated the initial training corpus from Tons of Inductive Problems; co-designed training losses exploiting proof context and Lean 4 verifier feedback.
Result: the **8B** model verifies **62.0% of 427 tasks** (*Verina*, *Clever*, *ALGOVERI*) — $2.6\times$ the strongest baseline, beating frontier reasoning models (18.5%) and neural provers up to $84\times$ larger.

Amazon Web Services, Santa Clara, CA & Boston, MA

Summer 2024, 2025, and 2026

Applied Scientist Intern, Agentic Automated Reasoning

- **SPECY** — specification learning for distributed systems (C#, Java; 2024)
Built: led the design and implementation — a specification grammar with quantifiers over events and a learner combining grammar-based enumerative search with dynamic learning from traces.
Result: learned **all 43 target specifications** across 11 open-source and 3 AWS service protocols (**28 fully automatically**, 15 with minor hints); recovered hand-written inductive invariants; exposed a test-coverage gap in a production AWS service. **First-authored at OOPSLA 2026.**
- **PVERIFIER** — deductive verification of P protocol models; shipped in the public release (2025)
Built: **Proof Script**, a DSL for proof decomposition in [PVERIFIER](#).
Result: proved safety and correctness properties (including consensus) for ring leader election, Paxos, and an internal AWS service.
- **Strata** — [foundational code verification framework](#) in Lean 4 (2026)
Built: the concurrency primitives (cooperative coroutines) to model async-Python programs; enabled compositional verification of *async-Python programs*

Taichi Graphics, Remote and Beijing, China

Jun 2022 – Sep 2022

Compiler R&D Intern

- **Taichi IR** — compiler optimization for tensor types
Built: a standalone **tensor type** in the Taichi IR; adapted the load/store forwarding, dead-code elimination, and reaching-definitions passes to handle tensor-typed expressions.

Intel, Strategic CAD Lab, Hillsboro, OR

Mar 2022 – Jun 2022

Formal Verification Research Intern

- **PYROPE** — correct-by-construction hardware modeling (Python, Dafny)
Built: a framework unifying a hardware model and its correctness proof, discharged by compiling to **Dafny**. *Result:* verified the **multi-Montgomery reduction** algorithm.

⚙️ TECHNICAL SKILLS

- **Formal Methods:** Lean 4, Dafny, P, Z3 / SMT solving, specification inference, equality saturation
- **LLMs & Agents:** Coding agents, vericoding (verified code generation), autoformalization, LLM-guided proof search, agent runtime enforcement
- **Programming:** Python, Java, C#, Rust · **Systems:** Distributed systems, compilers, program analysis

💡 TEACHING

- [COS 516: Automated Reasoning about Software](#) — TA, Princeton (Fall 2023 & 2024)
- [CSE 505: Principles of Programming Languages](#) — TA, University of Washington (Spring 2021)

🎤 SELECTED TALKS

- **SPECY: Learning Specifications for Distributed Systems** — *Invited*, Georgia Tech (Apr 2026)
- **Ranking Formal Specifications using LLMs** — SPLASH / LMPL'25 (Oct 2025)
- **Improving Term Extraction with Acyclic Constraints** — PLDI EGRAPHS'23 (Jun 2023)
- **Automated Super-Optimization for Taichi via Equality Saturation** — Taichi Graphics (Jan 2023)

🏆 HONORS & SERVICE

- CRA Outstanding Undergraduate Researcher Award, Honorable Mention (2022)
- Spotlight Papers: **ICML 2026** (×2), **ICLR 2021**
- Reviewer: NeurIPS, IEEE Transactions on Mobile Computing, AAE@KDD'25, SciPy'25
- Artifact Evaluation Committee: POPL'25, PLDI'24, POPL'24, MLSys'23, MICRO'21